



SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CIRCUIT DEBUGGING ASSESSMENT

ECA0405 – ANALOG CIRCUITS | CO3 | ASSESSMENT – 2

Student Name	G.Asmitha	Register No.	192512136
Class / Section	Analog circuits/ECA0405	Date	20/08/2026
Faculty Name	P.Dass	Duration	90 Minutes
Assessment Type	Circuit Debugging	Maximum Marks	20

Assessment Objective:

To identify, analyze, and correct errors in an analog circuit, justify the debugging approach, verify the corrected circuit operation, and relate the result to CO3

Question:

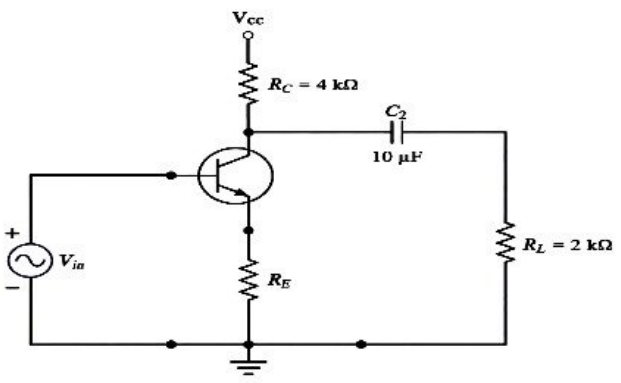
CO3 : Examine the frequency response characteristics of BJT and FET amplifiers at low and high frequencies to determine bandwidth, gain variation, and overall amplifier performance

CO3 AT2 Circuit Debugging:

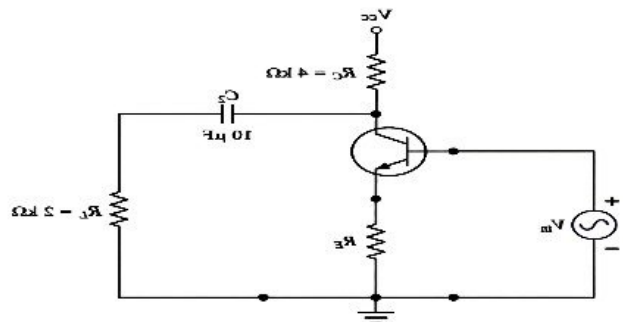
Description:

- Each student will be assigned one Circuit debugging problem 20 marks inside class. Analyze and debug one faulty analog circuit.
- Simulate the corrected circuit using LTspice and verify results.
- Document the debugging process, observations, and conclusions.
- Upload LTspice files, simulation results, and report to GitHub.
- Evaluation is based on fault identification, debugging approach, simulation accuracy, documentation, and GitHub submission.

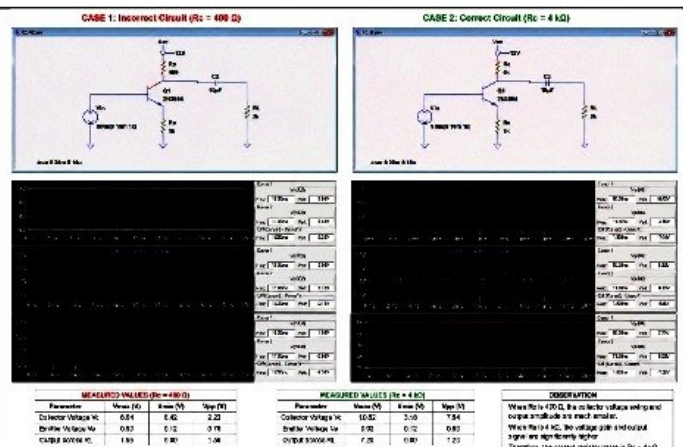
Circuit Debugging Record

Item	Student Entry
1. Given Circuit / Circuit Diagram	Paste or draw the assigned analog circuit here. 
2. Expected Circuit Operation	The circuit operates as a common-emitter BJT amplifier. The correct collector resistor is $R_C = 4 \text{ k}\Omega$. It provides proper collector voltage swing and voltage gain.
3. Observed Fault / Symptom	The collector resistor was mistakenly replaced by 400Ω instead of $4 \text{ k}\Omega$. This reduces the collector voltage swing and amplifier gain, causing a smaller output waveform.
4. Suspected Component / Connection	Collector resistor (R_C) has the wrong value: 400Ω instead of $4 \text{ k}\Omega$.
5. Debugging Analysis / Calculation	The resistor value is reduced by a factor of 10: $4 \text{ k}\Omega \rightarrow 400 \Omega$ Since the collector voltage variation depends on the voltage drop across R_C, $\Delta V_C = I_C \times R_C$ the smaller R_C produces a much smaller collector voltage swing. Therefore, the voltage gain and output amplitude decrease.
6. Error Identified	$R_C = 400 \Omega$ was used instead of the specified $R_C = 4 \text{ k}\Omega$.
7. Correction Made	Replace the 400Ω collector resistor with $4 \text{ k}\Omega$ and repeat the transient simulation.

8. Corrected Circuit Diagram



9. Verification / Simulation Output



10. Final Result & Technical Justification

The incorrect 400 Ω collector resistor reduces the collector voltage swing and voltage gain, resulting in a smaller output signal. Replacing it with the specified 4 kΩ resistor restores the intended amplifier operation. Therefore, $R_C = 4 \text{ k}\Omega$ is the appropriate resistor value.



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Marks Distribution

Assessment Criterion	Marks	Marks Awarded	Faculty Remarks
Problem identification & circuit understanding	4		
Fault analysis & technical calculations	5		
Correct identification of error	3		
Effective circuit correction	3		
Verification / output validation	3		
Technical explanation & documentation	2		
Total	20		